

ECE 501: Digital Image Processing

Course Project – Week 4 Progress Report

Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

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Project 17, Group 7

Project Title

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Team Details

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1 Work Done This Week

1.1 First Worked Example: Real Detection Results

The headline addition this week is a first end-to-end worked example. A Mahalanobis anomaly model was fit on normal-texture patches from 10 defect-free images (stride 64, 2,560 patches), using a per-fabric model where a fabric ID has its own defect-free images and a pooled (cross-fabric) model otherwise. Three real defective images were scored end to end (heatmap generation, thresholding at the 95th percentile of pooled normal scores (9.547), and evaluation against ground truth), giving the results in Table 1.

Table 1: Localization metrics, worked example (pooled model)

Image	IoU	Precision	Recall
0099_019_02	0.000	0.000	0.000
0102_010_03	0.248	0.281	0.679
0087_030_02	0.000	0.000	1.000
Mean	0.083	0.094	0.560

All three images used the pooled fallback model, since none of their parsed fabric IDs had a defect-free counterpart among the sampled normal images. We treat these numbers explicitly as a baseline, not a conclusion, given the sample size of three.

1.2 Failure-Mode Analysis

We analyzed why the three results differ so widely. Image 0087_030_02 (recall 1.0, precision near zero) had almost the entire image flagged as anomalous; image 0099_019_02 is a missed defect (the predicted mask had no overlap with ground truth); image 0102_010_03 is the best case (IoU 0.248). Our working hypothesis is that the pooled model mixes unrelated fabric textures into one normal-texture distribution, so a single global threshold ends up too loose for some fabrics and too tight for others. This is our leading candidate explanation for mean recall exceeding mean precision (0.560 vs. 0.094) and directly motivates per-fabric thresholding (Section 2). A second, built-in factor is that a 95th-percentile threshold flags about 5% of normal patches by construction, which limits precision regardless of the model.

1.3 Fabric-ID Audit Finding

Reviewing the audit outputs this week, we noticed that the parsed fabric IDs number 20 among defect-free images and 106 among defective images, with 17 in common. AFID contains only seven fabric types, so the leading filename token we parse behaves like an image index rather than a fabric type. This also means the one-ID-per-defective-image pattern is an indexing artifact, and it explains why the per-fabric model could not be used in the worked example. Correcting the parser against the AFID naming convention is now the first item of next week’s work.

1.4 Tied Dataset-Audit Findings to Design Choices

Near-total overlap between normal and defective image intensities (Fig. 1b in the submitted report) justifies not relying on raw intensity for detection. Defects being small relative to a 64×64 patch explains the 1:149 patch-level imbalance and our choice of IoU, precision and recall over accuracy. The limited overlap between parsed fabric IDs (17 shared, of 20 defect-free and 106 defective IDs) is what motivated the pooled fallback model used above; as noted, this overlap is itself affected by the parsing issue.

2 Work To Be Done

- Correct the fabric-ID parser against the AFID naming convention and re-run the worked example with per-fabric models.
- Evaluate on the full defective set (103 mask-matched images) rather than the 3-image worked example.
- Fit and compare per-fabric thresholds against the single pooled threshold, to test the failure-mode hypothesis directly.
- Add morphological post-processing (opening/closing) to the predicted mask to reduce the over-triggering seen in image 0087_030_02.
- Ablate the four texture descriptors (GLCM, LBP, Gabor, FFT) to identify which contribute most to detection accuracy.
- Check the grayscale assumption explicitly, since some sampled images reported up to four channels.
- Map numeric defect codes to AITEX’s named defect categories.
- Prepare and rehearse the 10-minute mid-semester presentation (tentatively 12th October), built around the worked example as the central result.

3 Literature Review

Finalized for the mid-semester report: the AFID dataset paper [1], the GLCM [2] and LBP [3] texture-descriptor papers, and a Gabor-filter texture-feature reference [4]. No changes this week beyond what was submitted; a texture-feature-on-AITEX benchmarking reference remains a candidate addition before end-semester.

4 Challenges Faced

- A three-image worked example is enough to demonstrate that the pipeline runs end to end, but is too small to draw real conclusions from; we framed it as a baseline rather than overstate it.
- The pooled model’s single global threshold appears to be the main driver of the precision/recall imbalance (0.094 vs. 0.560), but with three data points we cannot yet rule out other explanations (patch size, descriptor choice, the 95th-percentile threshold itself). Full-set evaluation should clarify this.
- The fabric-ID parsing issue (parsed IDs far exceed the seven AFID fabric types) means per-fabric modeling has not yet been properly tested; fixing it is a prerequisite for the next experiments.
- Fitting the required IEEE two-column, 2-page format meant compressing five pipeline stages, a dataset audit and a worked example into a tight space; earlier drafts had float-placement issues that needed explicit placement (`[!t]`, `\FloatBarrier`).

References

- [1] J. Silvestre-Blanes, T. Albero-Albero, I. Miralles, R. Pérez-Llorens, and J. Moreno, “A public fabric database for defect detection methods and results,” *Autex Research Journal*, vol. 19, no. 4, pp. 363–374, 2019.
- [2] R. M. Haralick, K. Shanmugam, and I. Dinstein, “Textural features for image classification,” *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.
- [3] T. Ojala, M. Pietikäinen, and T. Mäenpää, “Multiresolution gray-scale and rotation invariant texture classification with local binary patterns,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 24, no. 7, pp. 971–987, 2002.
- [4] B. S. Manjunath and W. Y. Ma, “Texture features for browsing and retrieval of image data,” *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 18, no. 8, pp. 837–842, Aug. 1996.